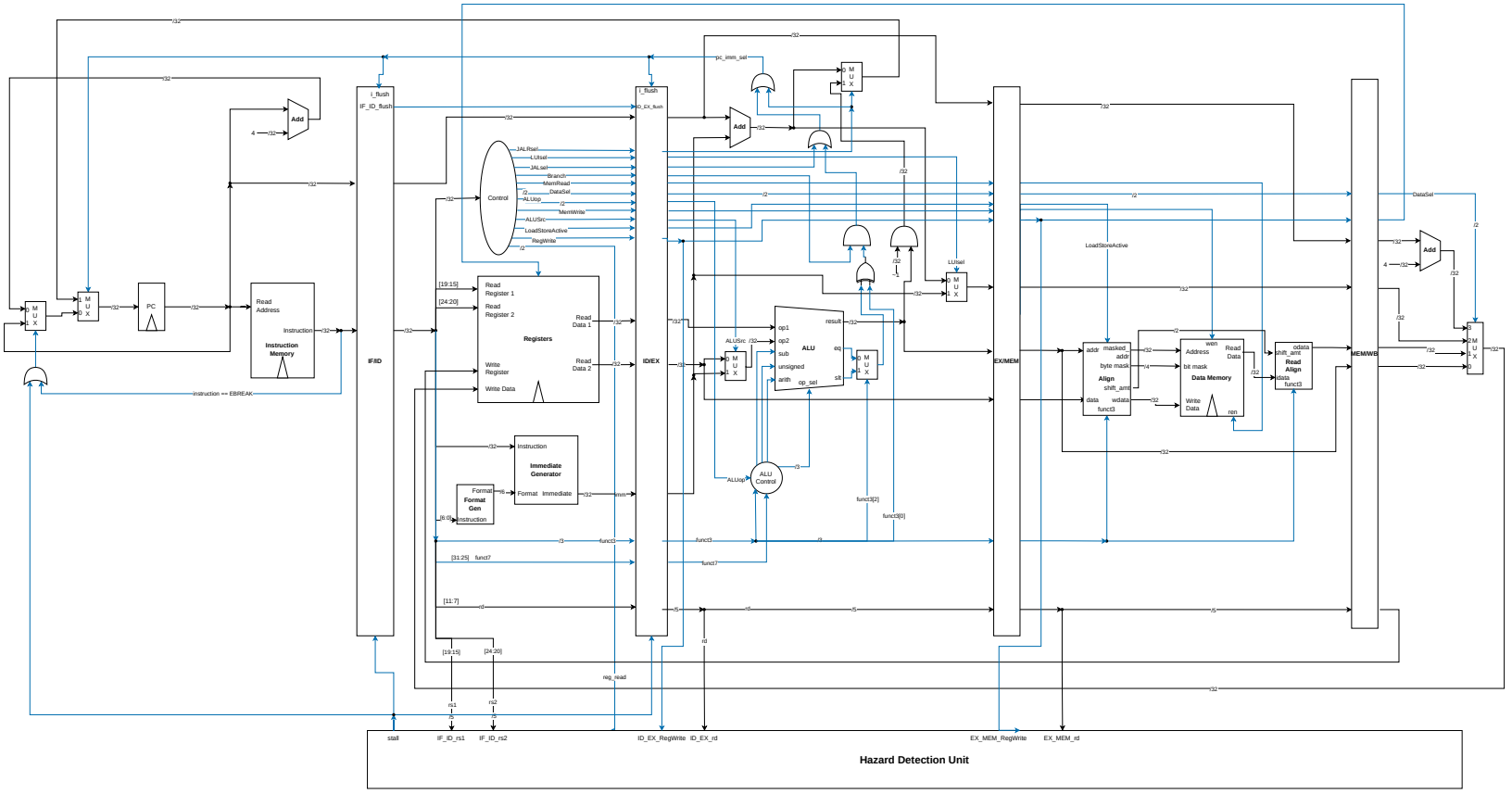

Add
4
PC
Read Address
Instruction
Instruction Memory
Instruction == EBREAK
i_flush
IF_ID_flush
IF/ID
Control
ALUSel
LUISel
JALSel
Branch
MemRead
DataSel
ALUop
MemWrite
ALUSrc
LoadStoreActive
RegWrite
Registers
Read Register 1
Read Register 2
Write Register
Write Data
Read Data 1
Read Data 2
[19:15]
[24:20]
Immediate Generator
Format Gen
Format
Immediate
[6:0]
[31:25] funct7
[11:7]
[19:15]
[24:20]
rs1
rs2
rd
funct3
funct7
ID/EX
ALU
op1
op2
sub
unsigned
arith
op_sel
result
eq
slt
ALUSrc
ALUop
ALU Control
funct3[2]
funct3[0]
pc_imm_sel
LUISel
EX/MEM
LoadStoreActive
Align
addr
masked addr
byte mask
shift_amt
data
wdata
funct3
Data Memory
Address
Read Data
Write Data
wen
ren
odata
shift_amt
Read Align
funct3
MEM/WB
DataSel
stall
IF_ID_rs1
IF_ID_rs2
ID_EX_RegWrite
ID_EX_rd
EX_MEM_RegWrite
EX_MEM_rd
Hazard Detection Unit